CHAPTER 6

Simple Regression Analysis

6.1 THE TWO-VARIABLE LINEAR MODEL

The two-variable linear model, or *simple regression analysis*, is used for testing hypotheses about the relationship between a dependent variable Y and an independent or explanatory variable X and for prediction. Simple *linear* regression analysis usually begins by plotting the set of XY values on a *scatter diagram* and determining by inspection if there exists an approximate linear relationship:

$$Y_i = b_0 + b_1 X_i \tag{6.1}$$

Since the points are unlikely to fall precisely on the line, the exact linear relationship in Eq. (*6.1*) must be modified to include a *random disturbance*, *error*, or *stochastic term*, u_i (see Sec. 1.2 and Prob. 1.8):

$$Y_i = b_0 + b_1 X_i + u_i \tag{6.2}$$

The error term is assumed to be (1) normally distributed, with (2) zero expected value or mean, and (3) constant variance, and it is further assumed (4) that the error terms are uncorrelated or unrelated to each other, and (5) that the explanatory variable assumes fixed values in repeated sampling (so that X_i and u_i are also uncorrelated).

EXAMPLE 1. Table 6.1 gives the bushels of corn per acre, Y, resulting from the use of various amounts of fertilizer in pounds per acre, X, produced on a farm in each of 10 years from 1971 to 1980. These are plotted in the scatter diagram of Fig. 6-1. The relationship between X and Y in Fig. 6-1 is approximately linear (i.e., the points would fall on or near a straight line).

6.2 THE ORDINARY LEAST-SQUARES METHOD

The *ordinary least-squares method* (OLS) is a technique for fitting the "best" straight line to the sample of XY observations. It involves minimizing the sum of the squared (vertical) deviations of points from the line:

$$\text{Min} \sum (Y_i - \hat{Y}_i)^2 \tag{6.3}$$

where Y_i refers to the actual observations, and $\hat{Y}_i$ refers to the corresponding *fitted* values, so that $Y_i - \hat{Y}_i = e_i$, the *residual*. This gives the following two *normal equations* (see Prob. 6.6):

Table 6.1 Corn Produced with Fertilizer Used

Year	n	Y_i	X_i
1971	1	40	6
1972	2	44	10
1973	3	46	12
1974	4	48	14
1975	5	52	16
1976	6	58	18
1977	7	60	22
1978	8	68	24
1979	9	74	26
1980	10	80	32

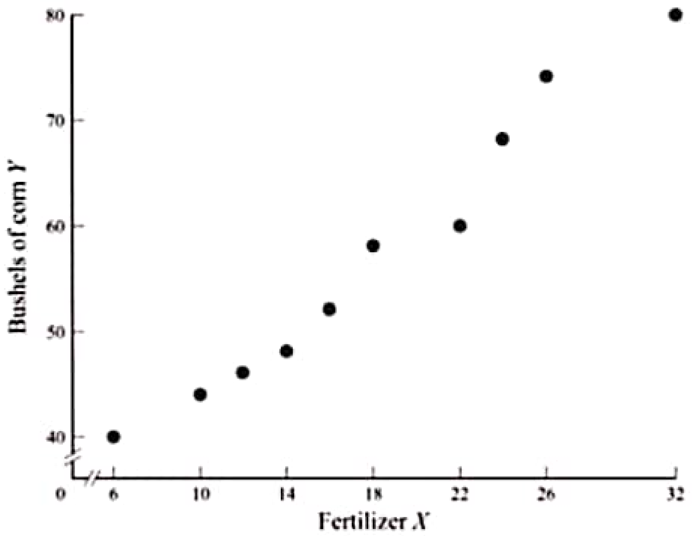

Fig. 6-1

$$\sum Y_i = n\hat{b}_0 + \hat{b}_1 \sum X_i \tag{6.4}$$

$$\sum X_i Y_i = \hat{b}_0 \sum X_i + \hat{b}_1 \sum X_i^2 \tag{6.5}$$

where n is the number of observations and $\hat{b}_0$ and $\hat{b}_1$ are estimators of the true parameters b_0 and b_1.

Solving simultaneously Eqs. (*6.4*) and (*6.5*), we get [see Prob. 6.7(*a*)]

$$\hat{b}_1 = \frac{n \sum X_i Y_i - \sum X_i \sum Y_i}{n \sum X_i^2 - (\sum X_i)^2} \tag{6.6}$$

The value of $\hat{b}_0$ is then given by [see Prob. 6.7(*b*)]

$$\hat{b}_0 = \overline{Y} - \hat{b}_1 \overline{X} \tag{6.7}$$

It is often useful to use an equivalent formula for estimating $\hat{b}_1$ [see Prob. 6.10(*a*)]:

$$\hat{b}_1 = \frac{\sum x_i y_i}{\sum x_i^2} = \frac{\text{cov}(X, Y)}{\sigma_X^2} \tag{6.8}$$

where $x_i = X_i - \overline{X}$, and $y_i = Y_i - \overline{Y}$. The estimated least-squares regression (OLS) equation is then

$$\hat{Y}_i = \hat{b}_0 + \hat{b}_1 X_i \tag{6.9}$$

EXAMPLE 2. Table 6.2 shows the calculations to estimate the regression equation for the corn-fertilizer problem in Table 6.1. Using Eq. (6.8),

Table 6.2 Corn Produced with Fertilizer Used: Calculations

n	Y_i (Corn)	X_i (Fertilizer)	y_i	x_i	$x_i y_i$	x_i^2
1	40	6	−17	−12	204	144
2	44	10	−13	−8	104	64
3	46	12	−11	−6	66	36
4	48	14	−9	−4	36	16
5	52	16	−5	−2	10	4
6	58	18	1	0	0	0
7	60	22	3	4	12	16
8	68	24	11	6	66	36
9	74	26	17	8	136	64
10	80	32	23	14	322	196
$n = 10$	$\sum Y_i = 570$ $\bar{Y} = 57$	$\sum X_i = 180$ $\bar{X} = 18$	$\sum y_i = 0$	$\sum x_i = 0$	$\sum x_i y_i = 956$	$\sum x_i^2 = 576$

$$\hat{b}_i = \frac{\sum x_i y_i}{\sum x_i^2} = \frac{956}{576} = 1.66 \qquad \text{(the slope of the estimated regression line)}$$

$$\hat{b}_0 = \bar{Y} - \hat{b}_1 \bar{X} \cong 57 - (1.66)(18) \cong 57 - 29.88 \cong 27.12 \qquad \text{(the } Y \text{ intercept)}$$

$$\hat{Y}_i = 27.12 + 1.66X_i \qquad \text{(the estimated regression equation)}$$

Thus, when $X_i = 0$, $\hat{Y} = 27.12 = \hat{b}_0$. When $X_i = 18 = \bar{X}$, $\hat{Y} = 27.12 + 1.66(18) = 57 = \bar{Y}$. As a result, the regression line passes through point $\overline{XY}$ (see Fig. 6-2).

6.3 TESTS OF SIGNIFICANCE OF PARAMETER ESTIMATES

In order to test for the statistical significance of the parameter estimates of the regression, the variance of $\hat{b}_0$ and $\hat{b}_1$ is required (see Probs. 6.14 and 6.15):

$$\operatorname{Var} \hat{b}_0 = \sigma_u^2 \frac{\sum X_i^2}{n \sum x_i^2} \tag{6.10}$$

$$\operatorname{Var} \hat{b}_1 = \sigma_u^2 \frac{1}{\sum x_i^2} \tag{6.11}$$

Since σ_u^2 is unknown, the *residual variance* s^2 is used as an (unbiased) estimate of σ_u^2:

$$s^2 = \hat{\sigma}_u^2 = \frac{\sum e_i^2}{n-k} \tag{6.12}$$

where k represents the number of parameter estimates.

Unbiased estimates of the variance of $\hat{b}_0$ and $\hat{b}_1$ are then given by

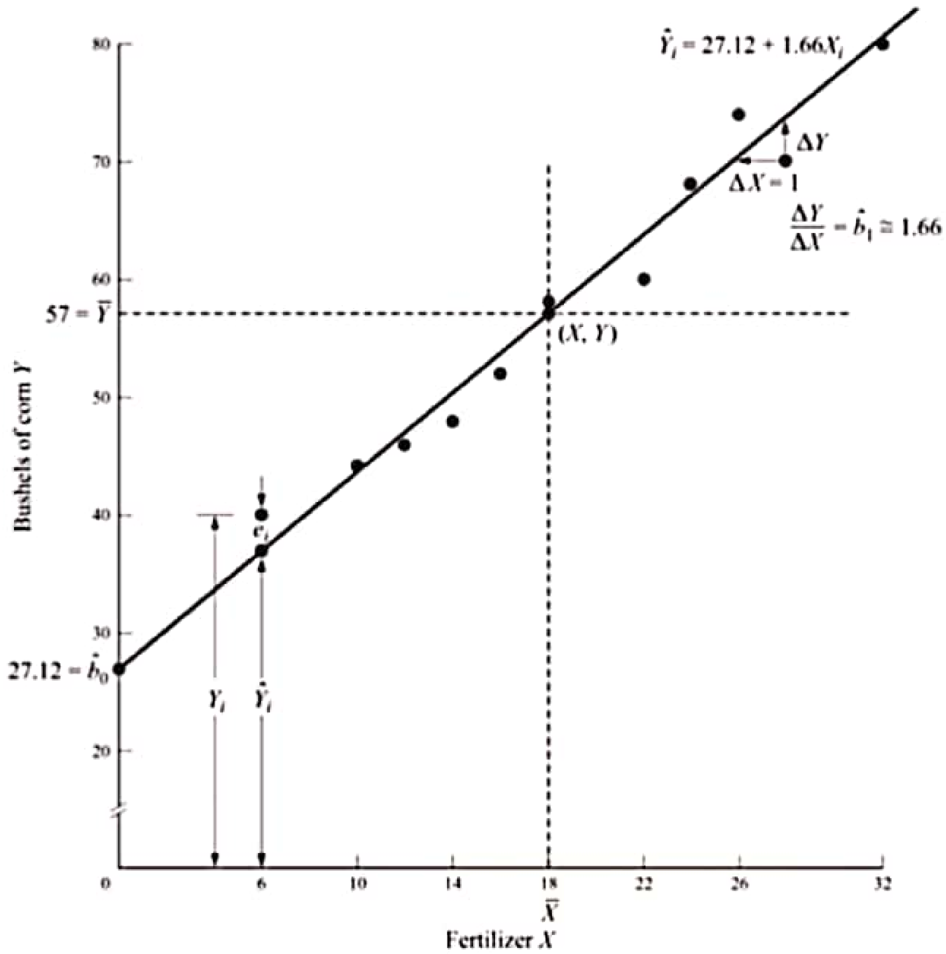

Fig. 6-2

$$s^2_{\hat{b}_0} = \frac{\sum e_i^2}{n-k} \frac{\sum X_i^2}{n \sum x_i^2} \tag{6.13}$$

$$s^2_{\hat{b}_1} = \frac{\sum e_i^2}{n-k} \frac{1}{\sum x_i^2} \tag{6.14}$$

so that $s_{\hat{b}_0}$ and $s_{\hat{b}_1}$ are the *standard errors of the estimates.* Since u_i is normally distributed, Y_i and therefore $\hat{b}_0$ and $\hat{b}_1$ are also normally distributed, so that we can use the t distribution with $n-k$ degrees of freedom, to test hypotheses about and construct confidence intervals for $\hat{b}_0$ and $\hat{b}_1$ (see Secs. 4.4 and 5.2).

EXAMPLE 3. Table 6.3 (an extension of Table 6.2) shows the calculations required to test the statistical significance of $\hat{b}_0$ and $\hat{b}_1$. The values of $\hat{Y}_i$ in Table 6.3 are obtained by substituting the values of X_i into the estimated regression equation found in Example 2. (The values of y_i^2 are obtained by squaring y_i from Table 6.2 and are to be used in Sec. 6.4.)

$$s^2_{\hat{b}_0} = \frac{\sum e_i^2}{n-k} \frac{\sum X_i^2}{n \sum x_i^2} \cong \frac{47.3056}{10-2} \frac{3816}{10(576)} \cong 3.92 \qquad \text{and} \qquad s_{\hat{b}_0} = \sqrt{3.92} \cong 1.98$$

$$s^2_{\hat{b}_1} = \frac{\sum e_i^2}{(n-k)\sum x_i^2} \cong \frac{47.3056}{(10-2)576} \cong 0.01 \qquad \text{and} \qquad s_{\hat{b}_1} \cong \sqrt{0.01} \cong 0.1$$

Therefore
$$t_0 = \frac{\hat{b}_0 - b_0}{s_{\hat{b}_0}} \cong \frac{27.12 - 0}{1.98} \cong 13.7 \qquad \text{and} \qquad t_1 = \frac{\hat{b}_1 - b_1}{s_{\hat{b}_1}} \cong \frac{1.66}{0.1} \cong 16.6$$

Table 6.3 Corn-Fertilizer Calculations to Test Significance of Parameters

Year	Y_i	X_i	$\hat{Y}_i$	e_i	e_i^2	X_i^2	x_i^2	y_i^2
1	40	6	37.08	2.92	8.5264	36	144	289
2	44	10	43.72	0.28	0.0784	100	64	169
3	46	12	47.04	−1.04	1.0816	144	36	121
4	48	14	50.36	−2.36	5.5696	196	16	81
5	52	16	53.68	−1.68	2.8224	256	4	25
6	58	18	57.00	1.00	1.0000	324	0	1
7	60	22	63.64	−3.64	13.2496	484	16	9
8	68	24	66.96	1.04	1.0816	576	36	121
9	74	26	70.28	3.72	13.8384	676	64	289
10	80	32	80.24	−0.24	0.0576	1024	196	529
$n = 10$				$\sum e_i = 0$	$\sum e_i^2 = 47.3056$	$\sum X_i^2 = 3816$	$\sum x_i^2 = 576$	$\sum y_i^2 = 1634$

Since both t_0 and t_1 exceed $t = 2.306$ with 8 df at the 5% level of significance (from App. 5), we conclude that both b_0 and b_1 are statistically significant at the 5% level.

6.4 TEST OF GOODNESS OF FIT AND CORRELATION

The closer the observations fall to the regression line (i.e., the smaller the residuals), the greater is the variation in Y "explained" by the estimated regression equation. The total variation in Y is equal to the explained plus the residual variation:

$$\underset{\substack{\text{Total variation in}\\ Y\text{ [or total sum of}\\ \text{squares}\\ \text{(TSS)]}}}{\sum(Y_i - \overline{Y})^2} = \underset{\substack{\text{Explained variation}\\ \text{in } Y \text{ [or regression}\\ \text{sum of squares}\\ \text{(RSS)]}}}{\sum(\hat{Y}_i - \overline{Y})^2} + \underset{\substack{\text{Residual variation}\\ \text{in } Y\text{ [or error sum}\\ \text{of squares}\\ \text{(ESS)]}}}{\sum(Y_i - \hat{Y}_i)^2} \tag{6.15}$$

Dividing both sides by TSS gives

$$1 = \frac{\text{RSS}}{\text{TSS}} + \frac{\text{ESS}}{\text{TSS}}$$

The *coefficient of determination*, or R^2, is then defined as the proportion of the total variation in Y "explained" by the regression of Y on X:

$$R^2 = \frac{\text{RSS}}{\text{TSS}} = 1 - \frac{\text{ESS}}{\text{TSS}} \tag{6.16}$$

R^2 can be calculated by

$$R^2 = \frac{\sum \hat{y}^2}{\sum y_i^2} = 1 - \frac{\sum e_i^2}{\sum y_i^2} \tag{6.17}$$

where
$$\sum \hat{y}_i^2 = \sum(\hat{Y}_i - \overline{Y}_i)^2$$

R^2 ranges in value from 0 (when the estimated regression equation explains none of the variation in Y) to 1 (when all points lie on the regression line).

The *correlation coefficient* r is given by (see Prob. 6.22)

$$r = \sqrt{R^2} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} = \sqrt{\hat{b}_1 \frac{\sum x_i y_i}{\sum y_i^2}} \tag{6.18}$$

r ranges in value from -1 (for perfect negative linear correlation) to $+1$ (for perfect positive linear correlation) and does not imply causality or dependence. With qualitative data, the *rank* or (the *Spearman*) *correlation coefficient* r' (see Prob. 6.25) can be used.

EXAMPLE 4. The coefficient of determination for the corn-fertilizer example can be found from Table 6.3:

$$R^2 = 1 - \frac{\sum e_i^2}{\sum y_i^2} \cong 1 - \frac{47.31}{1634} \cong 1 - 0.0290 \cong 0.9710, \text{ or } 97.10\%$$

Thus the regression equation explains about 97% of the total variation in corn output. The remaining 3% is attributed to factors included in the error term. Then $r = \sqrt{R^2} \cong \sqrt{0.9710} \cong 0.9854$, or 98.54%, and is positive because $\hat{b}_1$ is positive. Figure 6-3 shows the total, the explained, and the residual variation of Y.

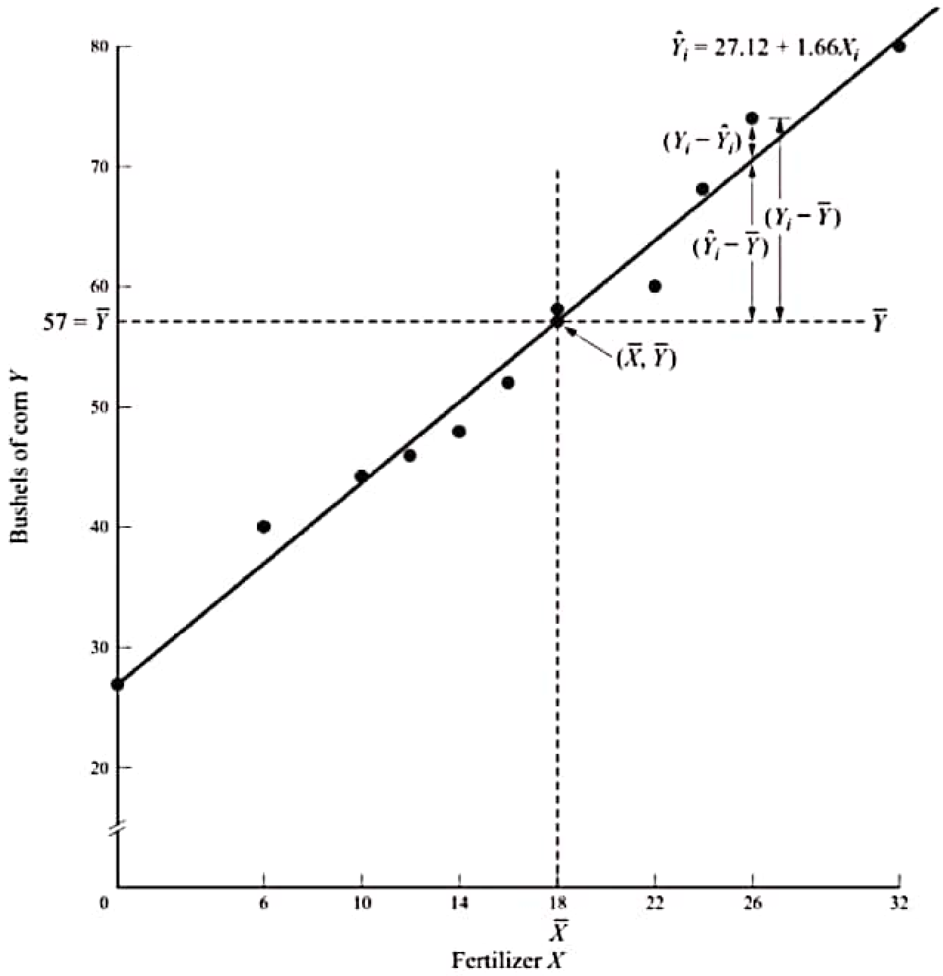

Fig. 6-3

6.5 PROPERTIES OF ORDINARY LEAST-SQUARES ESTIMATORS

Ordinary least-squares (OLS) estimators are *best linear unbiased estimators* (BLUE). Lack of bias means

$$E(\hat{b}) = b$$

so that

$$\text{Bias} = E(\hat{b}) - b$$

Best unbiased or efficient means smallest variance. Thus OLS estimators are the best among all unbiased linear estimators [see Probs. 6.14(a) and 6.15(b)]. This is known as the *Gauss-Markov theorem* and represents the most important justification for using OLS.

Sometimes, a researcher may want to trade off some bias for a possibly smaller variance and minimize the mean square error, MSE (see Prob. 6.29):

$$\text{MSE}(\hat{b}) = E(\hat{b} - b)^2 = \text{var}(\hat{b}) + (\text{bias}\,\hat{b})^2$$

An estimator is *consistent* if, as the sample size approaches infinity in the limit, its value approaches the true parameter (i.e., it is asymptotically unbiased) and its distribution collapses on the true parameter (see Prob. 6.30).

EXAMPLE 5. OLS estimators $\hat{b}_0$ and $\hat{b}_1$ found in Example 2 are unbiased linear estimators of b_0 and b_1 because

$$E(\hat{b}_0) = b_0 \qquad \text{and} \qquad E(\hat{b}_1) = b_1$$

Var $\hat{b}_0$ and var $\hat{b}_1$ found in Example 3 are also lower than for any other linear unbiased estimators. Therefore $\hat{b}_0$ and $\hat{b}_1$ are BLUE.

Solved Problems

THE TWO-VARIABLE LINEAR MODEL

6.1 What is meant by and what is the function of (*a*) Simple regression analysis? (*b*) Linear regression analysis? (*c*) A scatter diagram? (*d*) An error term?

(*a*) *Simple regression* is used for testing hypotheses about the relationship between a dependent variable Y and an independent or explanatory variable X and for prediction. This is to be contrasted with *multiple regression* analysis, in which there are not one, but two or more independent or explanatory variables. Multiple regression analysis is discussed in Chap. 7.

(*b*) *Linear regression analysis* assumes that there is an approximate linear relationship between X and Y (i.e., the set of random sample values of X and Y fall on or near a straight line). This is to be contrasted with *nonlinear regression analysis* (discussed in Sec. 8.1).

(*c*) A *scatter diagram* is a figure in which each pair of independent-dependent observations is plotted as a point in the XY plane. Its purpose is to determine (by inspection) if there exists an approximate linear relationship between the dependent variable Y and the independent or explanatory variable X.

(*d*) The *error term* (also known as the *disturbance* or *stochastic term*) measures the deviation of each observed Y value from the true (but unobserved) regression line. These error terms, designated by u_i and e_i, arise because of (1) numerous explanatory variables with only slight and irregular effects on Y that are omitted from the exact linear relationship given by Eq. (*6.1*), (2) possible errors of measurement in Y, and (3) random human behavior (see Prob. 1.8).

6.2 The data in Table 6.4 reports the aggregate consumption (Y, in billions of U.S. dollars) and disposable income (X, also in billions of U.S. dollars) for a developing economy for the 12 years from 1988 to 1999. Draw a scatter diagram for the data and determine by inspection if there exists an approximate linear relationship between Y and X.

From Fig. 6-4 it can be seen that the relationship between consumption expenditures Y and disposable income X is approximately linear, as required by the linear regression model.

6.3 State the general relationship between consumption Y and disposable income X in (*a*) exact linear form and (*b*) stochastic form. (*c*) Why would you expect most observed values of Y not to fall exactly on a straight line?

(*a*) The exact or deterministic general relationship between aggregate consumption expenditures Y and aggregate disposable income X can be written as

Table 6.4 Aggregate Consumption (Y) and Disposable Income (X)

Year	n	Y_i	X_i
1988	1	102	114
1989	2	106	118
1990	3	108	126
1991	4	110	130
1992	5	122	136
1993	6	124	140
1994	7	128	148
1995	8	130	156
1996	9	142	160
1997	10	148	164
1998	11	150	170
1999	12	154	178

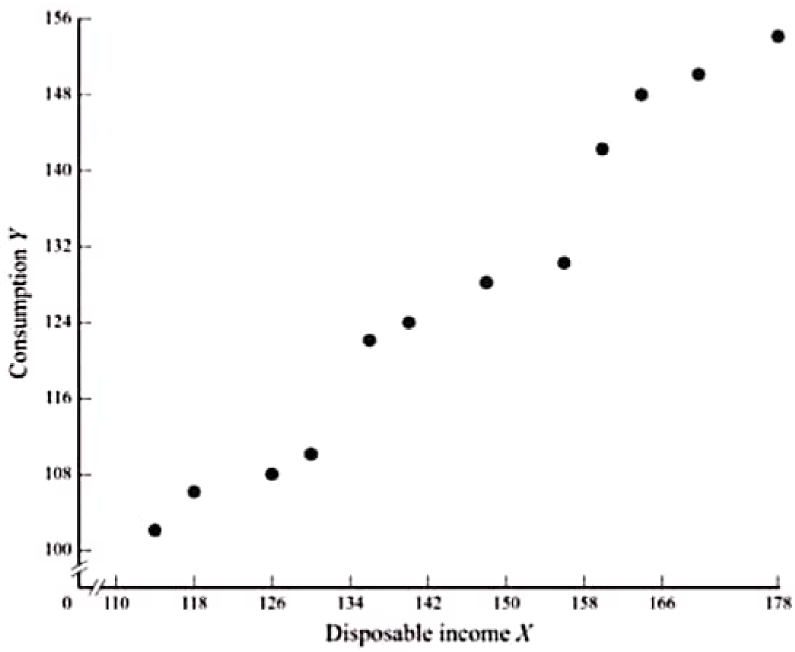

Fig. 6-4

$$Y_i = b_0 + b_1 X_i \tag{6.1}$$

where i refers to each year in time-series analysis (as with the data in Table 6.4) or to each economic unit (such as a family) in cross-sectional analysis. In Eq. (6.1), b_0 and b_1 are unknown constants called *parameters*. Parameter b_0 is the constant or Y intercept, while b_1 measures $\Delta Y/\Delta X$, which, in the context of Prob. 6.2, refers to the marginal propensity to consume (MPC) (see Sec. 1.2). The *specific* linear relationship corresponding to the general linear relationship in Eq. (*6.1*) is obtained by estimating the values of b_0 and b_1 (represented by $\hat{b}_0$ and $\hat{b}_1$ and read as "b sub zero hat" and "b sub one hat").

(*b*) The exact linear relationship in Eq. (*6.1*) can be made stochastic by adding a random disturbance or error term, u_i, giving

$$Y_i = b_0 + b_1 X_i + u_i \tag{6.2}$$

(*c*) Most observed values of Y are not expected to fall precisely on a straight line (1) because even though consumption Y is postulated to depend primarily on disposable income X, it also may depend on numerous other omitted variables with only slight and irregular effect on Y (if some of these other variables had instead a significant and regular effect on Y, then they should be included as additional explanatory variables, as in a multiple regression model); (2) because of possible errors in measuring Y; and (3) because of inherent random human behavior, which usually leads to different values of Y for the same value of X under identical circumstances (see Prob. 1.8).

6.4 State each of the five assumptions of the classical regression model (OLS) and give an intuitive explanation of the meaning and need for each of them.

1. The first assumption of the classical linear regression model (OLS) is that the random error term u is normally distributed. As a result, Y and the sampling distribution of the parameters of the regression are also normally distributed, so that tests can be conducted on the significance of the parameters (see Secs. 4.2, 5.2, and 6.3).

2. The second assumption is that the expected value of the error term or its mean equals zero:

$$E(u_i) = 0 \tag{6.19}$$

Because of this assumption, Eq. (*6.1*) gives the *average* value of Y. Specifically, since X is assumed fixed, the value of Y in Eq. (*6.2*) varies above and below its mean as u exceeds or is smaller than 0. Since the average value of u is assumed to be 0, Eq. (*6.1*) gives the average value of Y.

3. The third assumption is that the variance of the error term is constant in each period and for all values of X:

$$E(u_i)^2 = \sigma_u^2 \tag{6.20}$$

This assumption ensures that each observation is equally reliable, so that estimates of the regression coefficients are efficient and tests of hypotheses about them are not biased. These first three assumptions about the error term can be summarized as

$$u \sim N(0, \sigma_u^2) \tag{6.21}$$

4. The fourth assumption is that the value which the error term assumes in one period is uncorrelated or unrelated to its value in any other period:

$$E(u_i u_j) = 0 \qquad \text{for } i \neq j; \qquad i, j = 1, 2, \ldots, n \tag{6.22}$$

This ensures that the average value of Y depends only on X and not on u, and it is, once again, required in order to have efficient estimates of the regression coefficients and unbiased tests of their significance.

5. The fifth assumption is that the explanatory variable assumes fixed values that can be obtained in repeated samples, so that the explanatory variable is also uncorrelated with the error term:

$$E(X_i u_i) = 0 \tag{6.23}$$

This assumption is made to simplify the analysis.

THE ORDINARY LEAST-SQUARES METHOD

6.5 (*a*) What is meant by the *ordinary least-squares* (OLS) *method* of estimating the "best" straight line that fits the sample of XY observations? (*b*) Why do we take vertical deviations? (*c*) Why do we not simply take the sum of the deviations *without squaring them*? (*d*) Why do we not take the sum of the absolute deviations?

(*a*) The OLS method gives the best straight line that fits the sample of XY observations in the sense that it minimizes the sum of the squared (vertical) deviations of each observed point on the graph from the straight line.

(*b*) We take vertical deviations because we are trying to explain or predict movements in Y, which is measured along the vertical axis.

(*c*) We cannot take the sum of the deviations of each of the observed points from the OLS line because deviations that are equal in size but opposite in sign cancel out, so the sum of the deviations equals 0 (see Table 6.2).

(*d*) Taking the sum of the *absolute* deviations avoids the problem of having the sum of the deviations equal to 0. However, the sum of the *squared* deviations is preferred so as to penalize larger deviations relatively more than smaller deviations.

6.6 Starting from Eq. (*6.3*) calling for the minimization of the sum of the squared deviations or residuals, derive (*a*) normal Eq. (*6.4*) and (*b*) normal Eq. (*6.5*). (The reader without knowledge of calculus can skip this problem.)

(*a*)
$$\sum e_i^2 = \sum (Y_i - \hat{Y}_i)^2 = \sum (Y_i - \hat{b}_0 - \hat{b}_1 X_i)^2$$

Normal Eq. (*6.4*) is derived by minimizing $\sum e_i^2$ with respect to $\hat{b}_0$:

$$\frac{\partial \sum e_i^2}{\partial \hat{b}_0} = \frac{\partial \sum (Y_i - \hat{b}_0 - \hat{b}_1 X_i)^2}{\partial \hat{b}_0} = 0$$
$$2\sum (Y_i - \hat{b}_0 - \hat{b}_1 X_i)(-1) = 0$$
$$\sum (Y_i - \hat{b}_0 - \hat{b}_1 X_i) = 0$$
$$\sum Y_i = n\hat{b}_0 + \hat{b}_1 \sum X_i \tag{6.4}$$

(*b*) Normal Eq. (*6.5*) is derived by minimizing $\sum e_i^2$ with respect to $\hat{b}_1$:

$$\frac{\partial \sum e_i^2}{\partial \hat{b}_1} = \frac{\partial \sum (Y_i - \hat{b}_0 - \hat{b}_1 X_i)^2}{\partial \hat{b}_1} = 0$$
$$2\sum (Y_i - \hat{b}_0 - \hat{b}_1 X_i)(-X_i) = 0$$
$$\sum (Y_i X_i - \hat{b}_0 X_i - \hat{b}_1 X_i^2) = 0$$
$$\sum Y_i X_i = \hat{b}_0 \sum X_i + \hat{b}_1 \sum X_i^2 \tag{6.5}$$

6.7 Derive (*a*) Eq. (*6.6*) to find $\hat{b}_1$ and (*b*) Eq. (*6.7*) to find $\hat{b}_0$. [*Hint for part a*: Start by multiplying Eq. (*6.5*) by n and Eq. (*6.4*) by $\sum X_i$.]

(*a*) Multiplying Eq. (*6.5*) by n and Eq. (*6.4*) by $\sum X_i$, we get

$$n\sum X_i Y_i = \hat{b}_0 n \sum X_i + \hat{b}_1 n \sum X_i^2 \tag{6.24}$$
$$\sum X_i \sum Y_i = \hat{b}_0 n \sum X_i + \hat{b}_1 \left(\sum X_i\right)^2 \tag{6.25}$$

Subtracting Eq. (*6.25*) from Eq. (*6.24*), we get

$$n\sum X_i Y_i - \sum X_i \sum Y_i = \hat{b}_1 \left[n \sum X_i^2 - \left(\sum X_i\right)^2 \right] \tag{6.26}$$

Solving Eq. (*6.26*) for $\hat{b}_1$, we get

$$\hat{b}_1 = \frac{n\sum X_i Y_i - \sum X_i \sum Y_i}{n\sum X_i^2 - (\sum X_i)^2} \tag{6.6}$$

(*b*) Equation (*6.7*) is obtained by simply solving Eq. (*6.4*) for $\hat{b}_0$:

$$\sum Y_i = n\hat{b}_0 + \hat{b}_1 \sum X_i \tag{6.4}$$
$$\hat{b}_0 = \frac{\sum Y_i}{n} - \hat{b}_1 \frac{\sum X_i}{n}$$
$$= \bar{Y} - \hat{b}_1 \bar{X} \tag{6.7}$$

6.8 (*a*) State the difference between b_0 and b_1, on one hand, and $\hat{b}_0$ and $\hat{b}_1$ on the other hand. (*b*) State the difference between u_i and e_i. (*c*) Write the equations for the true and estimated *relationships* between X and Y. (*d*) Write the equations for the true and estimated *regression* lines between X and Y.

(*a*) b_0 and b_1 are the parameters of the true but unknown regression line, while $\hat{b}_0$ and $\hat{b}_1$ are the parameters of the estimated regression line.

(*b*) u_i is the random disturbance, error, or stochastic term in the true but unknown relationship between X and Y. However, e_i is the residual between each observed value of Y and its corresponding fitted value $\hat{Y}$ in the estimated relationship.

(*c*) The equations for the true and estimated relationships between X and Y are, respectively,

$$Y_i = b_0 + b_1 X_i + u_i \tag{6.2}$$

$$Y_i = \hat{b}_0 + \hat{b}_1 X_i + e_i \tag{6.27}$$

(*d*) The equations for the true and estimated regressions between X and Y are, respectively,

$$E(Y_i) = b_0 + b_1 X_i \tag{6.28}$$

$$\hat{Y}_i = \hat{b}_0 + \hat{b}_1 X_i \tag{6.9}$$

6.9 (*a*) Find the regression equation for the consumption schedule in Table 6.4, using Eq. (*6.6*) to find $\hat{b}_1$. (*b*) Plot the regression line and show the deviations of each Y_i from the corresponding $\hat{Y}_i$.

(*a*) Table 6.5 shows the calculations to find $\hat{b}_1$ and $\hat{b}_0$ for the data in Table 6.4.

$$\hat{b}_1 = \frac{n\sum X_i Y_i - \sum X_i \sum Y_i}{n\sum X_i^2 - (\sum X_i)^2} = \frac{(12)(225{,}124) - (1740)(1524)}{(12)(257{,}112) - (1740)^2} = \frac{2{,}701{,}488 - 2{,}651{,}760}{3{,}085{,}344 - 3{,}027{,}600}$$

$$= \frac{49{,}728}{57{,}744} \cong 0.86$$

$$\hat{b}_0 = \overline{Y} - \hat{b}_1 \overline{X} \cong 127 - 0.86(145) \cong 127 - 124.70 \cong 2.30$$

Thus the equation for the estimated consumption regression is $\hat{Y}_i = 2.30 + 0.86\hat{X}_i$.

Table 6.5 Aggregate Consumption and Disposable Income: Calculations

n	Y_i	X_i	$X_i Y_i$	X_i^2
1	102	114	11,628	12,996
2	106	118	12,508	13,924
3	108	126	13,608	15,876
4	110	130	14,300	16,900
5	122	136	16,592	18,496
6	124	140	17,360	19,600
7	128	148	18,944	21,904
8	130	156	20,280	24,336
9	142	160	22,720	25,600
10	148	164	24,272	26,896
11	150	170	25,500	28,900
12	154	178	27,412	31,684
$n = 12$	$\sum Y_i = 1524$ $\overline{Y} = 127$	$\sum X_i = 1740$ $\overline{X} = 145$	$\sum X_i Y_i = 225{,}124$	$\sum X_i^2 = 257{,}112$

(*b*) To plot the regression equation, we need to define any two points on the regression line. For example, when $X_i = 114$, $Y_i = 2.30 + 0.86(114) = 100.34$. When $X_i = 178$, $Y_i = 2.30 + 0.86(178) = 155.38$.

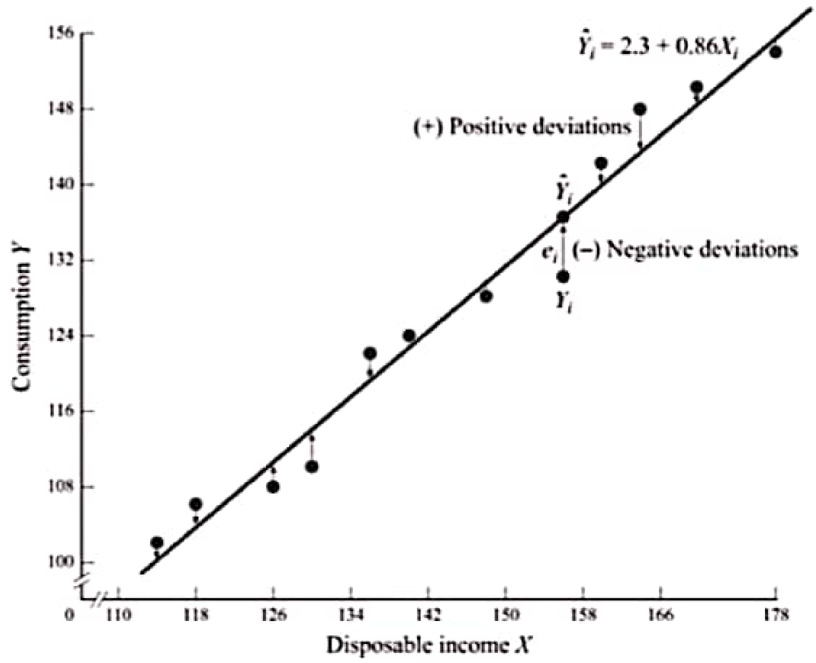

Fig. 6-5

The consumption regression line is plotted in Fig. 6-5, where the positive and negative residuals are also shown. The regression line represents the best fit to the random sample of consumption–disposable income observations in the sense that it minimizes the sum of the squared (vertical) deviations from the line.

6.10 (*a*) Starting with Eq. (*6.6*), derive the equation for $\hat{b}_1$ in deviation form for the case where $\overline{X} = \overline{Y} = 0$. (*b*) What is the value of $\hat{b}_0$ when $\overline{X} = \overline{Y} = 0$?

(*a*) Starting with Eq. (*6.6*) for $\hat{b}_1$

$$\hat{b}_1 = \frac{n\sum X_iY_i - \sum X_i \sum Y_i}{n\sum X_i^2 - (\sum X_i)^2} \tag{6.6}$$

we divide numerator and denominator by n^2 and get

$$\begin{aligned}
\hat{b} &= \frac{\sum X_iY_i/n - (\sum X_i/n)(\sum Y_i/n)}{\sum X_i^2/n - (\sum X_i/n)^2} \\
&= \frac{\sum X_iY_i/n - \overline{X}\,\overline{Y}}{\sum X_i^2/n - \overline{X}^2} \\
&= \frac{\sum X_iY_i}{\sum X_i^2} \qquad \text{since } \overline{X} = \overline{Y} = 0 \text{ and canceling the } n \text{ terms} \\
&= \frac{\sum x_iy_i}{\sum x_i^2} \qquad \text{since } \overline{X} = \overline{Y} = 0
\end{aligned} \tag{6.8}$$

(*b*) Starting with Eq. (*6.7*) for $\hat{b}_0$, we obtain

$$\hat{b}_0 = \overline{Y} - \hat{b}_1\overline{X} \tag{6.7}$$

and substituting 0 for $\overline{X}$ and $\overline{Y}$, we get

$$\hat{b}_0 = 0 - \hat{b}_1(0) = 0$$

6.11 With respect to the data in Table 6.4, (*a*) find the value of $\hat{b}_1$ using Eq. (*6.8*), and (*b*) plot the regression line on a graph measuring the variables as deviations from their respective means. How does this regression line compare with the regression line plotted in Fig. 6-5?

(*a*) Table 6.6 shows the calculations to find $\hat{b}_1$ for the data in Table 6.4. In deviation form (note that $\sum y_i = \sum x_i = 0$):

$$\hat{b}_1 = \frac{\sum x_i y_i}{\sum x_i^2} = \frac{4144}{4812} \cong 0.86 \qquad \text{[the same as in Prob. 6.9(}a\text{)]}$$

Table 6.6 Aggregate Consumption and Disposable Income: Alternative Calculations

n	Y_i	X_i	y_i	x_i	$x_i y_i$	x_i^2
1	102	114	−25	−31	775	961
2	106	118	−21	−27	567	729
3	108	126	−19	−19	361	361
4	110	130	−17	−15	255	225
5	122	136	−5	−9	45	81
6	124	140	−3	−5	15	25
7	128	148	1	3	3	9
8	130	156	3	11	33	121
9	142	160	15	15	225	225
10	148	164	21	19	399	361
11	150	170	23	25	575	625
12	154	178	27	33	891	1089
			$\sum y_i = 0$	$\sum x_i = 0$	$\sum x_i y_i = 4144$	$\sum x_i^2 = 4812$

(*b*) From Prob. 6.10(*b*) we know that the regression line crosses the origin when plotted on a graph with the axis measuring the variables in deviation form, and from part *a* of this problem we know that this regression line has the same slope as the regression line in Fig. 6-5. See Fig. 6-6.

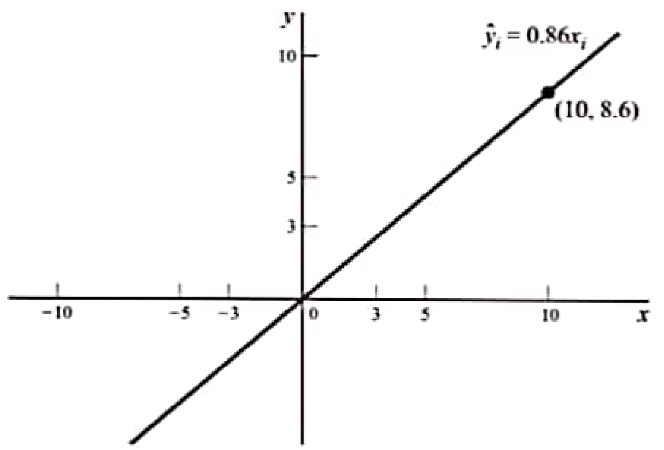

Fig. 6-6

6.12 In the context of Prob. 6.9(*a*), what is the meaning of: (*a*) Estimator $\hat{b}_0$? (*b*) Estimator $\hat{b}_1$? (*c*) Find the income elasticity of consumption.

(*a*) Estimator $\hat{b}_0 \cong 2.30$ is the Y intercept, or the value of aggregate consumption, in billions of dollars, when disposable income, also in billions of dollars, is 0. The fact that $\hat{b}_0 > 0$ confirms what was anticipated on theoretical grounds in Example 3 in Chap. 1.

(b) Estimator $\hat{b}_1 = dY/dX \cong 0.86$ is the slope of the estimated regression line. It measures the marginal propensity to consume (MPC) or the change in consumption per one-unit change in disposable income. Once again, the fact that $0 < \hat{b}_1 < 1$ confirms what was anticipated on theoretical grounds in Example 3 in Chap. 1.

(c) The income elasticity of consumption η measures the percentage change in consumption resulting from a given percentage change in disposable income. Since the elasticity usually changes at every point in the function, it is measured at the means:

$$\eta = \hat{b}_1 \frac{\overline{X}}{\overline{Y}} \tag{6.29}$$

For the data in Table 6.4

$$\eta = 0.86 \frac{145}{127} \cong 0.98$$

Note that elasticity, as opposed to the slope, is a pure (unit-free) number.

TESTS OF SIGNIFICANCE OF PARAMETER ESTIMATES

6.13 Define (a) σ_u^2 and s^2, (b) $\operatorname{var}\hat{b}_0$ and $\operatorname{var}\hat{b}_1$, (c) $s_{\hat{b}_0}^2$ and $s_{\hat{b}_1}^2$, and (d) $s_{\hat{b}_0}$ and $s_{\hat{b}_1}$.

(a) σ_u^2 is the variance of the error term in the true relationship between X_i and Y_i. However, $s^2 = \hat{\sigma}_u^2 = \sum e_i^2/(n-k)$ is the residual variance and is an (unbiased) estimate of σ_u^2, which is unknown. k is the number of estimated parameters. In simple regression analysis, $k = 2$. Thus $n - k = n - 2$ and refers to the degrees of freedom.

(b) $\operatorname{Var}\hat{b}_0 = \sigma_u^2 \sum X_i/n\sum x_i^2$, while $\operatorname{var}\hat{b}_1 = \sigma_u^2/\sum x_i^2$. The variances of $\hat{b}_0$ and $\hat{b}_1$ (or their estimates) are required to test hypotheses about and construct confidence intervals for $\hat{b}_0$ and $\hat{b}_1$.

(c)
$$s_{\hat{b}_0}^2 = s^2 \frac{\sum X_i^2}{n\sum x_i^2} = \frac{\sum e_i^2 \sum X_i^2}{(n-k)n\sum x_i^2} \qquad \text{and} \qquad s_{\hat{b}_1}^2 = \frac{s^2}{\sum x_i^2} = \frac{\sum e_i^2}{(n-k)\sum x_i^2}$$

$s_{\hat{b}_0}^2$ and $s_{\hat{b}_1}^2$ are, respectively, (unbiased) estimates of $\operatorname{var}\hat{b}_0$ and $\operatorname{var}\hat{b}_1$, which are unknown since σ_u^2 is unknown.

(d) $s_{\hat{b}_0} = \sqrt{s_{\hat{b}_0}^2}$ and $s_{\hat{b}_1} = \sqrt{s_{\hat{b}_1}^2}$. $s_{\hat{b}_0}$ and $s_{\hat{b}_1}$ are, respectively, the standard deviations of $\hat{b}_0$ and $\hat{b}_1$ and are called the *standard errors*.

6.14 Prove that (a) mean $\hat{b}_1 = b_1$, and (b) $\operatorname{var}\hat{b}_1 = \sigma_u^2/\sum x_i^2$
(c) mean $\hat{b}_0 = b_0$, and (d) $\operatorname{var}\hat{b}_0 = \sigma_u^2(\sum X_i^2/n\sum x_i^2)$

(a)
$$\begin{aligned}\hat{b}_1 &= \frac{\sum x_i y_i}{\sum x_i^2} = \frac{\sum x_i(Y_i - \overline{Y})}{\sum x_i^2} = \frac{\sum x_i Y_i}{\sum x_i^2} - \frac{\sum x_i \overline{Y}}{\sum x_i^2} = \frac{\sum x_i Y_i}{\sum X_i^2} \qquad \text{since } \sum x_i = 0 \\ &= \sum c_i Y_i\end{aligned}$$

where $c_i = x_i/\sum x_i^2$ = constant because of assumption 5 (Sec. 6.1)

$$\begin{aligned}\hat{b}_1 &= \sum c_i Y_i = \sum c_i(b_0 + b_1 X_i + u_i) = b_0 \sum c_i + b_1 \sum c_i X_i + \sum c_i u_i \\ &= b_1 + \sum c_i u_i = b_1 + \frac{\sum x_i u_i}{\sum x_i^2}\end{aligned}$$

since $\sum c_i = \sum x_i/\sum x_i^2 = 0$ (because $\sum x_i = 0$) and

$$\sum c_i X_i = \frac{\sum x_i X_i}{\sum x_i^2} = \frac{\sum(X_i - \overline{X})X_i}{\sum(X_i - \overline{X})^2} = \frac{\sum X_i^2 - \overline{X}\sum X_i}{\sum X_i^2 - 2\overline{X}\sum X_i + n\overline{X}^2} = \frac{\sum X_i^2 - \overline{X}\sum X_i}{\sum X_i^2 - \overline{X}\sum X_i} = 1$$

$$E(\hat{b}_1) = E(b_1) + E\left[\frac{\sum x_i u_i}{\sum x_i^2}\right] = E(b_1) + \frac{1}{\sum x_i^2} E\left(\sum x_i u_i\right) = b_1$$

since b_1 is a constant and $E(\sum x_i u_i) = 0$ because of assumption 5 (Sec. 6.1).

(*b*) From part *a*, we obtain

$$b_1 = \frac{\sum x_i Y_i}{\sum x_i^2} = \sum c_i Y_i$$

$$\operatorname{Var} \hat{b}_1 = \operatorname{var}\left(\sum c_i Y_i\right) = \sum c_i^2 \operatorname{var} Y_i = \sum c_i^2 \sigma_u^2$$

since Y_i varies only because of u_i with X_i assumed fixed.

$$\operatorname{Var} \hat{b}_1 = \sum c_i^2 \sigma_u^2 = \sum\left(\frac{x_i}{\sum x_i^2}\right)^2 \sigma_u^2 = \frac{\sum x_i^2}{\left(\sum x_i^2\right)^2}\sigma_u^2 = \frac{\sigma_u^2}{\sum x_i^2}$$

(*c*)

$$\hat{b}_0 = \overline{Y} - \hat{b}_1\overline{X} = \frac{\sum Y_i}{n} - \overline{X}\sum c_i Y_i \qquad \text{(from part } a\text{)}$$

$$\hat{b}_0 = \frac{\sum Y_i}{n} - \overline{X}\sum c_i Y_i = \sum\left(\frac{1}{n} - \overline{X}c_i\right) Y_i$$

$$E(b_0) = \sum\left(\frac{1}{n} - \overline{X}c_i\right) E(Y_i) = \sum\left(\frac{1}{n} - \overline{X}c_i\right)(b_0 + b_1 X_i) \qquad \text{[from Eq. (6.1) in Prob. 6.8(}d\text{)]}$$

Cross multiplying,

$$E(b_0) = \sum\left(\frac{b_0}{n} - \overline{X}c_i b_0 + \frac{b_1 X_i}{n} - \overline{X}c_i b_1 X_i\right) = b_0 + b_1\overline{X} - b_1\overline{X} = b_0$$

because $\sum c_i = 0$ and $\sum c_i X_i = 1$, from part *a*.

(*d*) We saw in part *c* that

$$\hat{b}_0 = \sum\left(\frac{1}{n} - \overline{X}c_i\right) Y_i$$

$$\operatorname{Var} \hat{b}_0 = \operatorname{var}\left[\sum\left(\frac{1}{n} - \overline{X}c_i\right) Y_i\right] = \sum\left(\frac{1}{n} - \overline{X}c_i\right)^2 \operatorname{var} Y_i = \sigma_u^2 \sum\left(\frac{1}{n} - \overline{X}c_i\right)^2$$

$$\operatorname{Var} \hat{b}_0 = \sigma_u^2 \sum\left(\frac{1}{n^2} - \frac{2\overline{X}c_i}{n} + \overline{X}^2 c_i^2\right) = \sigma_u^2\left(\frac{1}{n} + \frac{\overline{X}^2}{\sum x_i^2}\right) = \sigma_u^2 \frac{\sum x_i^2 + n\overline{X}^2}{n\sum x_i^2}$$

since $\sum c_i = 0$ and $\sum c_i^2 = 1/\sum x_i^2$.

$$\operatorname{Var} \hat{b}_0 = \sigma_u^2 \frac{\sum x_i^2 + n\overline{X}^2}{n\sum x_i^2} = \sigma_u^2 \frac{\sum X_i^2 - n\overline{X}^2 + n\overline{X}^2}{n\sum x_i^2} = \sigma_u^2 \frac{\sum X_i^2}{n\sum x_i^2}$$

since in part *a* we saw that $\sum x_i^2 = \sum X_i^2 - n\overline{X}^2$.

6.15 For the aggregate consumption-income observations in Table 6.4, find (*a*) s^2, (*b*) $s_{\hat{b}_0}^2$ and $s_{\hat{b}_0}$, (*c*) $s_{\hat{b}_1}^2$ and $s_{\hat{b}_1}$.

(*a*) The calculations required to find s^2 are shown in Table 6.7, which is an extension of Table 6.6. The values for $\hat{Y}_i$ in Table 6.7 are obtained by substituting the values for X_i into the regression equation found in Prob. 6.9(*a*).

$$s^2 = \hat{\sigma}_u^2 = \frac{\sum e_i^2}{n-k} = \frac{115.2752}{12-2} = 11.52752 \cong 11.53$$

(*b*)

$$s_{\hat{b}_0}^2 = \frac{\sum e_i^2}{n-k}\frac{\sum X_i^2}{n\sum x_i^2} = \frac{s\sum X_i^2}{n\sum x_i^2} \cong \frac{(11.53)(257{,}112)}{(12)(4812)} \cong 51.34$$

Then

$$s_{\hat{b}_0} = \sqrt{s_{\hat{b}_0}^2} \cong \sqrt{51.34} \cong 7.17$$

Table 6.7 Consumption Regression: Calculations to Test Significance of Parameters

Year	Y_i	X_i	$\hat{Y}_i$	e_i	e_i^2	X_i^2	x_i^2
1	102	114	100.34	1.66	2.7556	12,996	961
2	106	118	103.78	2.22	4.9284	13,924	729
3	108	126	110.66	−2.66	7.0756	15,876	361
4	110	130	114.10	−4.10	16.8100	16,900	225
5	122	136	119.26	2.74	7.5076	18,496	81
6	124	140	122.70	1.30	1.6900	19,600	25
7	128	148	129.58	−1.58	2.4964	21,904	9
8	130	156	136.46	−6.46	41.7316	24,336	121
9	142	160	139.90	2.10	4.4100	25,600	225
10	148	164	143.34	4.66	21.7156	26,896	361
11	150	170	148.50	1.50	2.2500	28,900	625
12	154	178	155.38	−1.38	1.9044	31,684	1089
				$\sum e_i = 0$	$\sum e_i^2 = 115.2752$	$\sum X_i^2 = 27{,}112$	$\sum x_i^2 = 4812$

(*c*)
$$s_{\hat{b}_1}^2 = \frac{\sum e_i^2}{(n-k)\sum x_i^2} = \frac{s^2}{\sum x_i^2} \cong \frac{11.53}{4812} \cong 0.0024$$

Then
$$s_{\hat{b}_1} = \sqrt{s_{\hat{b}_1}^2} \cong \sqrt{0.0024} \cong 0.05$$

6.16 (*a*) State the null and alternative hypotheses to test the statistical significance of the parameters of the regression equation estimated in Prob. 6.9(*a*). (*b*) What is the form of the sampling distribution of $\hat{b}_0$ and $\hat{b}_1$? (*c*) Which distribution must we use to test the statistical significance of b_0 and b_1? (*d*) What are the degrees of freedom?

(*a*) To test for the statistical significance of b_0 and b_1, we set the following null hypothesis, H_0, and alternative hypothesis, H_1 (see Sec. 5.2):

$$H_0\colon\ b_0 = 0 \quad \text{versus} \quad H_1\colon\ b_0 \neq 0$$
$$H_0\colon\ b_1 = 0 \quad \text{versus} \quad H_1\colon\ b_1 \neq 0$$

The hope in regression analysis is to reject H_0 and to accept H_1, that b_0 and $b_1 \neq 0$, with a two-tail test.

(*b*) Since u_i is assumed to be normally distributed (assumption 1 in Sec. 6.1), Y_i is also normally distributed (since X_i is assumed to be fixed—assumption 5). As a result, $\hat{b}_0$ and $\hat{b}_1$ also will be normally distributed.

(*c*) To test the statistical significance of b_0 and b_1, the t distribution (from App. 5) must be used because $\hat{b}_0$ and $\hat{b}_1$ are normally distributed, but var $\hat{b}_0$ and var $\hat{b}_1$ are unknown (since σ_u^2 is unknown) and $n < 30$ (see Sec. 4.4).

(*d*) The degrees of freedom are $n - k$, where n is the number of observations and k is the number of parameters estimated. Since in simple regression analysis, two parameters are estimated ($\hat{b}_0$ and $\hat{b}_1$), df $= n - k = n - 2$.

6.17 Test at the 5% level of significance for (*a*) b_0 and (*b*) b_1 in Prob. 6.9(*a*).

(*a*)
$$t_0 = \frac{\hat{b}_0 - b_0}{s_{\hat{b}_0}} \cong \frac{2.30 - 0}{7.17} \cong 0.32$$

Since t_0 is smaller than the tabular value of $t = 2.228$ at the 5% level (two-tail test) and with 10 df (from App. 5), we conclude that b_0 is not statistically significant at the 5% level (i.e., we cannot reject H_0, that $b_0 = 0$).

(*b*)
$$t_1 = \frac{\hat{b}_1 - b_1}{s_{\hat{b}_1}} \cong \frac{0.86 - 0}{0.05} \cong 17.2$$

So b_1 is statistically significant at the 5% (and 1%) level (i.e., we cannot reject H_1, that $b_1 \neq 0$).

6.18 Construct the 95% confidence interval for (*a*) b_0 and (*b*) b_1 in Prob. 6.9(*a*).

(*a*) The 95% confidence interval for b_0 is given by (Sec. 4.4)

$$b_0 = \hat{b}_0 \pm 2.228 s_{\hat{b}_0} = 2.30 \pm 2.228(7.17) = 2.30 \pm 15.97$$

So b_0 is between -13.67 and 18.27 with 95% confidence. Note how wide (and meaningless) the 95% confidence interval b_0 is, reflecting the fact that $\hat{b}_0$ is highly insignificant.

(*b*) The 95% confidence interval for b_1 is given by

$$b_1 = \hat{b}_1 \pm 2.228 s_{\hat{b}_1} = 0.86 \pm 2.228(0.05) = 0.86 \pm 0.11$$

So b_1 is between 0.75 and 0.97 (i.e., $0.75 < b_1 < 0.97$) with 95% confidence.

TEST OF GOODNESS OF FIT AND CORRELATION

6.19 Derive the formula for R^2.

The coefficient of determination R^2 is defined as the proportion of the total variation in Y "explained" by the regression of Y on X. The total variation in Y or total sum of squares TSS $= \sum(Y_i - \overline{Y})^2 = \sum y_i^2$. The explained variation in Y or regression sum of squares RSS $= \sum(\hat{Y}_i - \overline{Y})^2 = \sum \hat{y}_i^2$. The residual variation in Y or error sum of squares ESS $= \sum(Y_i - \hat{Y}_i)^2 = \sum e_i^2$.

$$\begin{aligned} \text{TSS} &= \text{RSS} &&+ \text{ESS} \\ \sum(Y_i - \overline{Y})^2 &= \sum(\hat{Y}_i - \overline{Y})^2 &&+ \sum(Y_i - \hat{Y}_i)^2 \\ \sum y_i^2 &= \sum \hat{y}_i^2 &&+ \sum e_i^2 \end{aligned}$$

Dividing both sides by $\sum y_i^2$, we get

$$\frac{\sum y_i^2}{\sum y_i^2} = \frac{\sum \hat{y}_i^2}{\sum y_i^2} + \frac{\sum e_i^2}{\sum y_i^2}$$

$$1 = \frac{\sum \hat{y}_i^2}{\sum y_i^2} + \frac{\sum e_i^2}{\sum y_i^2}$$

Therefore
$$R^2 = \frac{\sum \hat{y}_i^2}{\sum y_i^2} = 1 - \frac{\sum e_i^2}{\sum y_i^2}$$

R^2 is unit-free and $0 \leq R^2 \leq 1$ because $0 \leq \text{ESS} \leq \text{TSS}$. $R^2 = 0$ when, for example, all sample points lie on a horizontal line $Y = \overline{Y}$ or on a circle. $R^2 = 1$ when all sample points lie on the estimated regression line, indicating a perfect fit.

6.20 (*a*) What does the correlation coefficient measure? What is its range of values? (*b*) What is the relationship between correlation and regression analysis?

(*a*) The correlation coefficient measures the degree of association between two or more variables. In the two-variable case, the simple linear correlation coefficient, r, for a set of sample observations is given by

$$r = \sqrt{R^2} = \frac{\sum x_i y_i}{\sqrt{\sum x_i^2}\sqrt{\sum y_i^2}} = \sqrt{\hat{b}_1 \frac{\sum x_i y_i}{\sum y_i^2}}$$

$-1 \le r \le +1$. $r < 0$ means that X and Y move in opposite directions, such as, for example, the quantity demanded of a commodity and its price. $r > 0$ indicates that X and Y change in the same direction, such as the quantity supplied of a commodity and its price. $r = -1$ refers to a perfect negative correlation (i.e., all the sample observations lie on a straight line of negative slope); however, $r = 1$ refers to perfect positive correlation (i.e., all the sample observations lie on a straight line of positive slope). $r = \pm 1$ is seldom found. The closer r is to ± 1, the greater is the degree of positive or negative linear relationship. It should be noted that the sign of r is always the same as that of $\hat{b}_1$. A zero correlation coefficient means that there exists no linear relationship whatsoever between X and Y (i.e., they tend to change with no connection with each other). For example, if the sample observations fall exactly on a circle, there is a perfect nonlinear relationship but a zero linear relationship, and $r = 0$.

(*b*) Regression analysis implies (but does not prove) causality between the independent variable X and dependent variable Y. However, correlation analysis implies no causality or dependence but refers simply to the type and degree of association between two variables. For example, X and Y may be highly correlated because of another variable that strongly affects both. Thus correlation analysis is a much less powerful tool than regression analysis and is seldom used by itself in the real world. In fact, the main use of correlation analysis is to determine the degree of association found in regression analysis. This is given by the coefficient of determination, which is the square of the correlation coefficient.

6.21 Derive the equation (*a*) $r = \sum x_i y_i / (\sqrt{\sum x_i^2}\sqrt{\sum y_i^2})$ (*Hint*: Start by showing that $\sum x_i y_i$ is a measure of association between X and Y.) and (*b*) $r = \sqrt{\hat{b}_1(\sum x_i y_i / \sum y_i^2)}$ [*Hint*: Start with $r = \sum x_i y_i / (\sqrt{\sum x_i^2}\sqrt{\sum y_i^2})$.]

(*a*) $\sum x_i y_i$ provides a measure of the association between X and Y because if X and Y both rise or fall, $x_i y_i > 0$, while if X rises and Y falls, or vice versa, $x_i y_i < 0$. If all or most sample observations involve a rise or fall in both X and Y, $\sum x_i y_i > 0$ and large, implying a large positive correlation. If all or most sample observations involve opposite changes in X and Y, then $\sum x_i y_i < 0$ and large, implying a large negative correlation. If, however, some X and Y observations move in the same direction, while others move in opposite directions, $\sum x_i y_i$ will be smaller, indicating a small net positive or negative correlation. However, measuring the degree of association by $\sum x_i y_i$ has two disadvantages. First, the greater is the number of sample observations, the larger is $\sum x_i y_i$; and second, $\sum x_i y_i$ is expressed in the units of the problem. These problems can be overcome by dividing $\sum x_i y_i$ by n (the number of sample observations) and by the standard deviation of X and Y ($\sqrt{\sum x_i^2/n}$ and $\sqrt{\sum y_i^2/n}$). Then

$$\frac{\sum x_i y_i}{n} = \text{covariance of } X \text{ and } Y \tag{6.30}$$

and

$$\frac{\sum x_i y_i}{n\sqrt{\dfrac{\sum x_i^2}{n}}\sqrt{\dfrac{\sum y_i^2}{n}}} = \frac{\sum x_i y_i}{\sqrt{\sum x_i^2}\sqrt{\sum y_i^2}} = r$$

(*b*)

$$r = \frac{\sum x_i y_i}{\sqrt{\sum x_i^2}\sqrt{\sum y_i^2}} = \frac{\sqrt{\sum x_i y_i}}{\sqrt{\sum x_i^2}}\,\frac{\sqrt{\sum x_i y_i}}{\sqrt{\sum y_i^2}} = \sqrt{\hat{b}_1 \frac{\sum x_i y_i}{\sum y_i^2}}$$

6.22 Find R^2 for the estimated consumption regression of Prob. 6.9 using the equation (*a*) $R^2 = \sum \hat{y}_i^2 / \sum y_i^2$ and (*b*) $R^2 = 1 - \sum e_i^2 / \sum y_i^2$.

(*a*) From Prob. 6.19, we know that $\sum y_i^2 = \sum \hat{y}_i^2 + \sum e_i^2$, so $\sum \hat{y}_i^2 = \sum y_i^2 - \sum e_i^2$. Since $\sum y_i^2 = 3684$ (by squaring and adding the y_i terms from Table 6.6) and $\sum e_i^2 = 115.2572$ (from Table 6.7) $\sum \hat{y}_i^2 = 3684 - 115.2572 = 3568.7428$. Thus

$$R^2 = \frac{\sum \hat{y}_i^2}{\sum y_i^2} = \frac{3568.7428}{3684} \cong 0.9687, \text{ or } 96.87\%$$

(*b*) Using $\sum e_i^2 = 115.2572$ and $\sum y_i^2 = 3684$, we get

$$R^2 = 1 - \frac{\sum e_i^2}{\sum y_i^2} = 1 - \frac{115.2572}{3684} \cong 0.9687, \text{ or } 96.87\%$$

(as in part *a*).

6.23 Find r for the estimated consumption regression in Prob. 6.9 using (*a*) $\sqrt{R^2}$, (*b*) $r = \sum x_i y_i / (\sqrt{\sum x_i^2}\sqrt{\sum y_i^2})$, and (*c*) $r = \sqrt{\hat{b}_1(\sum x_i y_i / \sum y_i^2)}$.

(*a*) $r = \sqrt{R^2} \cong \sqrt{0.9687} \cong 0.9842$ and is positive because $\hat{b}_1 > 0$.

(*b*) Using $\sum x_i y_i = 4144$ and $\sum x_i^2 = 4812$ from Table 6.6 and $\sum y_i^2 = 3684$ from Prob. 6.22(*a*), we get

$$r = \frac{\sum x_i y_i}{\sqrt{\sum x_i^2}\sqrt{\sum y_i^2}} \cong \frac{4144}{\sqrt{4812}\sqrt{3684}} \cong 0.9841$$

The very small difference between the value of r found here and that found in part *a* results from rounding errors.

(*c*) Using $\hat{b}_1 \cong 0.86$ found in Prob. 6.9(*a*), we obtain

$$r = \sqrt{\hat{b}_1 \frac{\sum x_i y_i}{\sum y_i^2}} \cong \sqrt{\frac{(0.86)(4144)}{3684}} \cong 0.9836$$

6.24 (*a*) Find the rank or Spearman correlation coefficient between the midterm grade and the IQ ranking of a random sample of 10 students in a large class, as given in Table 6.8, using Eq. (*6.31*). (*b*) When is the rank correlation used?

Table 6.8 Midterm Grade and IQ Ranking

Student	1	2	3	4	5	6	7	8	9	10
Midterm grade	77	78	65	84	84	88	67	92	68	96
IQ ranking	7	6	8	5	4	3	9	1	10	2

(*a*)
$$r' = 1 - \frac{6\sum D^2}{n(n^2 - 1)} \tag{6.31}$$

where D is the difference between ranks of corresponding pairs of the two variables (either in ascending or descending order, with the mean rank assigned to observations of the same value) and n is the number of observations.

The calculations to find r' are given in Table 6.9.

$$r' = 1 - \frac{6\sum D^2}{n(n^2 - 1)} = 1 - \frac{6(10.50)}{10(99)} = 1 - \frac{63}{990} \cong 0.94$$

(*b*) Rank correlation is used with qualitative data such as profession, education, or sex, when, because of the absence of numerical values, the coefficient of correlation cannot be found. Rank correlation also is used when precise values for all or some of the variables are not available (so that, once again, the coefficient of correlation cannot be found). Furthermore, with a great number of observations of large values, r' can be found as an estimate of r in order to avoid very time-consuming calculations (however, easy accessibility to computers has practically eliminated this reason for using r').

Table 6.9 Calculations to Find the Coefficient of Rank Correlation

n	Midterm Grade	Ranking on Midterm	IQ Ranking	D	D^2
1	96	1	2	−1	1
2	92	2	1	1	1
3	88	3	3	0	0
4	84	4.5	4	0.5	0.25
5	84	4.5	5	−0.5	0.25
6	78	6	6	0	0
7	77	7	7	0	0
8	68	8	10	−2	4
9	67	9	9	0	0
10	65	10	8	2	4
					$\sum D^2 = 10.50$

PROPERTIES OF ORDINARY LEAST-SQUARES ESTIMATORS

6.25 (*a*) What is meant by an unbiased estimator? How is bias defined? (*b*) Draw a figure showing the sampling distribution of an unbiased and a biased estimator.

(*a*) An estimator is *unbiased* if the mean of its sampling distribution equals the true parameter. The mean of the sampling distribution is the expected value of the estimator. Thus *lack of bias* means that $E(\hat{b}) = b$, where $\hat{b}$ is the estimator of the true parameter, b. *Bias* is then defined as the difference between the expected value of the estimator and the true parameter; that is, bias $= E(\hat{b}) - b$. Note that lack of bias does not mean that $\hat{b} = b$, but that in repeated random sampling, we get, on average, the correct estimate. The hope is that the sample actually obtained is close to the mean of the sampling distribution of the estimator.

(*b*) Figure 6-7*a* shows the sampling distribution of an estimator that is unbiased, and Fig. 6-7*b* shows one that is biased.

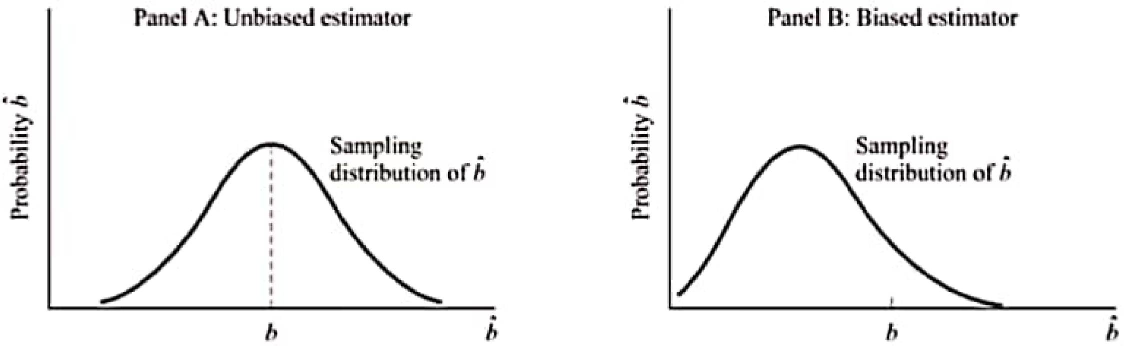

Fig. 6-7

6.26 (*a*) What is meant by the best unbiased or efficient estimator? Why is this important? (*b*) Draw a figure of the sampling distribution of two unbiased estimators, one of which is efficient.

(*a*) The *best unbiased or efficient estimator* refers to the one with the smallest variance among unbiased estimators. It is the unbiased estimator with the most compact or least spread-out distribution. This is very important because the researcher would be more certain that the estimator is closer to the true population parameter being estimated. Another way of saying this is that an efficient estimator has the smallest confidence interval and is more likely to be statistically significant than any other estimator. It should be noted that minimum variance by itself is not very important, unless coupled with the lack of bias.

(*b*) Figure 6-8*a* shows the sampling distribution of an efficient estimator, while Fig. 6-8*b* shows an inefficient estimator.

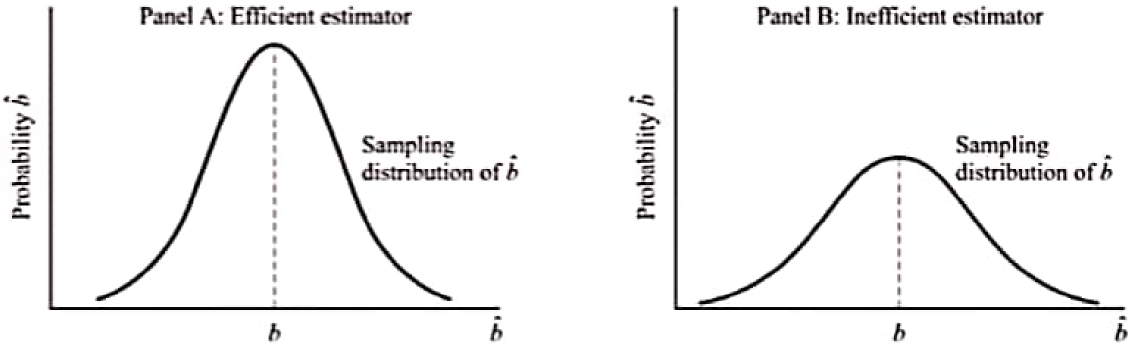

Fig. 6-8

6.27 Why is the OLS estimator so widely used? Is it superior to all other estimators?

The OLS estimator is widely used because it is BLUE (best linear unbiased estimator). That is, among all unbiased linear estimators, it has the lowest variance. The BLUE properties of the OLS estimator is often referred to as the *Gauss-Markov theorem*. However, *nonlinear* estimators may be superior to the OLS estimator (i.e., they might be unbiased and have lower variance). Since it is often difficult or impossible to find the variance of unbiased nonlinear estimators, however, the OLS estimator remains by far the most widely used. The OLS estimator, being linear, is also easier to use than nonlinear estimators.

6.28 (*a*) What is meant by the *mean-square error*? Why and when is the rule to minimize the mean-square error useful? (*b*) Prove that the mean-square error equals the variance plus the square of the bias of the estimator.

(*a*)
$$\text{MSE}(\hat{b}) = E(\hat{b} - b)^2 = \text{var}\,\hat{b} + (\text{bias}\,\hat{b})^2$$

The rule to minimize the MSE arises when the researcher faces a slightly biased estimator but with a smaller variance than any unbiased estimator. The researcher is then likely to choose the estimator with the lowest MSE. This rule penalizes equally for the larger variance or for the square of the bias of an estimator. However, this is used only when the OLS estimator has an "unacceptably large" variance.

(*b*)
$$\begin{aligned}\text{MSE}(\hat{b}) &= E(\hat{b} - b)^2\\ &= E[\hat{b} - E(\hat{b}) + E(\hat{b}) - b]^2\\ &= E[\hat{b} - E(\hat{b})]^2 + [E(\hat{b}) - b]^2 + 2E\{[\hat{b} - E(\hat{b})][E(\hat{b}) - b]\}\\ &= \text{var}\,\hat{b} + (\text{bias}\,\hat{b})^2\end{aligned}$$

because $E[\hat{b} - E(\hat{b})]^2 = \text{var}\,\hat{b}$, $[E(\hat{b}) - b]^2 = (\text{bias}\,\hat{b})^2$, and $E\{[\hat{b} - E(\hat{b})][E(\hat{b}) - b]\} = 0$ because this expression is equal to $E\{\hat{b}E(\hat{b}) - [E(\hat{b})]^2 - \hat{b}b + bE(\hat{b})\} = [E(\hat{b})]^2 - [E(\hat{b})]^2 - bE(\hat{b}) + bE(\hat{b}) = 0$.

6.29 (*a*) What is meant by consistency? (*b*) Draw a figure of the sampling distribution of a consistent estimator.

(*a*) Two conditions are required for an estimator to be consistent: (1) as the sample size increases, the estimator must approach more and more the true parameter (this is referred to as *asymptotic unbiasedness*); and (2) as the sample size approaches infinity in the limit, the sampling distribution of the estimator must collapse or become a straight vertical line with height (probability) of 1 above the value of the true parameter. This large-sample property of consistency is used only in situations when small-sample BLUE or lowest MSE estimators cannot be found.

(*b*) In Fig. 6-9, $\hat{b}$ is a consistent estimator of b because as n increases, $\hat{b}$ approaches b, and as n approaches infinity in the limit, the sampling distribution of $\hat{b}$ collapses on b.

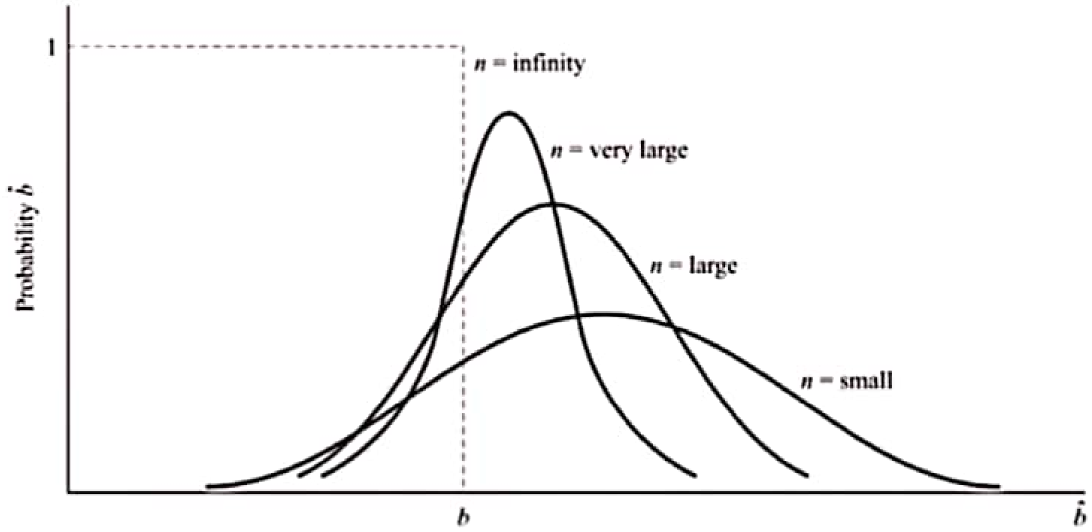

Fig. 6-9

SUMMARY PROBLEM

6.30 Table 6.10 gives the per capita income to the nearest \$100 (Y) and the percentage of the economy represented by agriculture (X) reported by the World Bank World Development Indicators for 1999 for 15 Latin American countries. (*a*) Estimate the regression equation of Y_i on X_i. (*b*) Test at the 5% level of significance for the statistical significance of the parameters. (*c*) Find the coefficient of determination. (*d*) Report the results obtained in part *a* in standard summary form.

Table 6.10 Per Capita Income (Y, \$00) and Percentage of the Economy in Agriculture (X)

Country:*	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Y_i	76	10	44	47	23	19	13	19	8	44	4	31	24	59	37
X_i	6	16	9	8	14	11	12	10	18	5	26	8	8	9	5

**Key*: (1) Argentina; (2) Bolivia; (3) Brazil; (4) Chile; (5) Colombia; (6) Dominican Republic; (7) Ecuador; (8) El Salvador; (9) Honduras; (10) Mexico; (11) Nicaragua; (12) Panama; (13) Peru; (14) Uruguay; (15) Venezuela.
Source: World Bank World Development Indicators.

(*a*) The first seven columns of Table 6.11 are used to answer part *a*. The rest of the table is filled by utilizing the results of part *a* in order to answer parts *b* and *c* of this problem.

$$\hat{b}_1 = \frac{\sum x_i y_i}{\sum x_i^2} = \frac{-1149}{442} \cong -2.60$$

$$\hat{b}_0 = \bar{Y} - \hat{b}_1 \bar{X} = 30.53 - (-2.60)(11) = 59.13$$

$$\hat{Y}_i = 59.13 - 2.60X_i$$

n	Y_i	X_i	y_i	x_i	$x_i y_i$	x_i^2	$\hat{Y}_i$	e_i	e_i^2	X_i^2	Y_i^2
1	76	6	45.47	−5	−227.35	25	43.53	32.47	1054.3009	36	2067.5209
2	10	16	−20.53	5	−102.65	25	17.53	−7.53	56.7009	256	421.4809
3	44	9	13.47	−2	−26.94	4	35.73	8.27	68.3929	81	181.4409
4	47	8	16.47	−3	−49.41	9	38.33	8.67	75.1689	64	271.2609
5	23	14	−7.53	3	−22.59	9	22.73	0.27	0.0729	196	56.7009
6	19	11	−11.53	0	0	0	30.53	−11.53	132.9409	121	132.9409
7	13	12	−17.53	1	−17.53	1	27.93	−14.93	222.9049	144	307.3009
8	19	11	−11.53	−1	11.53	1	33.13	−14.13	199.6569	100	132.9409
9	8	18	−22.53	7	−157.71	49	12.33	−4.33	18.7489	324	507.6009
10	44	5	13.47	−6	−80.82	36	46.13	−2.13	4.5369	25	181.4409
11	4	26	−26.53	15	−397.95	225	−8.47	12.47	155.5009	676	703.8409
12	31	8	0.47	−3	−1.41	9	38.33	−7.33	53.7289	64	0.2209
13	24	8	−6.53	−3	19.59	9	38.33	−14.33	205.3489	64	42.6409
14	59	9	28.47	−2	−56.94	4	35.73	23.27	541.4929	81	810.5409
15	37	5	6.47	−6	−38.82	36	46.13	−9.13	83.3569	25	41.9609
	$\sum Y_i = 458$ $\overline{Y} \cong 30.53$	$\sum X_i = 165$ $\overline{X} = 11.00$			$\sum y_i x_i = -1149$	$\sum x_i^2 = 442$			$\sum e_i^2 = 2872.8535$	$\sum X_i^2 = 2257$	$\sum y_i^2 = 5859.7335$

(*b*)
$$s^2_{\hat{b}_0} = \frac{\sum e_i^2}{(n-k)}\frac{\sum X_i^2}{n\sum x_i^2} = \frac{(2872.8535)(2257)}{(15-2)(15)(442)} \cong 75.23 \qquad \text{and} \qquad s_{\hat{b}_0} \cong 8.67$$

$$s^2_{\hat{b}_1} = \frac{\sum e_i^2}{(n-k)\sum x_i^2} = \frac{(2872.8535)}{(15-2)(442)} \cong 0.050 \qquad \text{and} \qquad s_{\hat{b}_1} = 0.71$$

$$t_0 = \frac{\hat{b}_0}{s^2_{\hat{b}_0}} = \frac{59.13}{8.67} \cong 6.82$$

$$t_1 = \frac{\hat{b}_1}{s^2_{\hat{b}_1}} = \frac{-2.60}{0.71} \cong -3.66$$

Therefore both $\hat{b}_0$ and $\hat{b}_1$ are statistically significant at the 5% level.

(*c*)
$$R^2 = 1 - \frac{\sum e_i^2}{\sum y_i^2} = 1 - \frac{2872.8535}{5859.7335} \cong 0.51$$

(*d*)
$$\hat{Y}_i = \underset{(6.82)}{59.13} - \underset{(-3.66)}{2.60}X_i \qquad R^2 = 0.51$$

The numbers in parentheses below the estimated parameters are the corresponding t values. An alternative way is to report the standard error of the estimates in parentheses.

Supplementary Problems

THE TWO-VARIABLE LINEAR MODEL

6.31 Draw a scatter diagram for the data in Table 6.12 and determine by inspection if there is an approximate linear relationship between Y_i and X_i.
Ans. The relationship between X and Y in Fig. 6-10 is approximately linear.

Table 6.12 Observations on Variables Y and X

n	Y_i	X_i
1	20	2
2	28	3
3	40	5
4	45	4
5	37	3
6	52	5
7	54	7
8	43	6
9	65	7
10	56	8

6.32 State the assumptions of the classical linear regression (OLS) model in mathematical form.
Ans.

$$u \sim N(0, \sigma_u^2) \tag{6.21}$$

$$E(u_i u_j) = 0 \qquad \text{for } i \neq j; \quad i, j = 1, 2, \ldots, n \tag{6.22}$$

$$E(X_i u_i) = 0 \tag{6.23}$$

(See Prob. 6.4.)

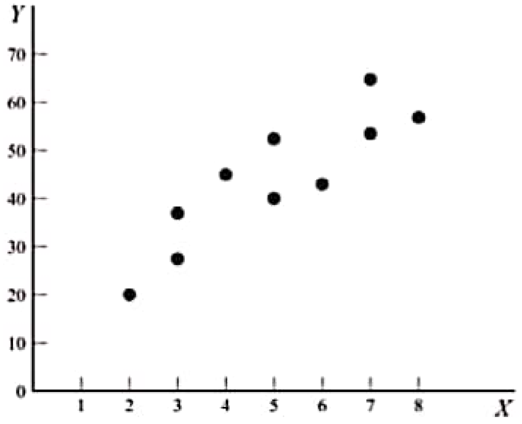

Fig. 6-10

THE ORDINARY LEAST-SQUARES METHOD

6.33 Express mathematically the following statements and formulas: (*a*) Minimize the sum of the squared deviations of each value of Y from its corresponding fitted value. (*b*) Minimize the sum of squared residuals. (*c*) The normal equations. (*d*) The formulas for estimating $\hat{b}_1$ and $\hat{b}_0$.
Ans. (*a*) Min $\sum (Y_i - \hat{Y}_i)^2$ (*b*) Min$\sum e_i^2$ (*c*) $\sum Y_i = n\hat{b}_0 + \hat{b}_1 \sum X_i$ and $\sum X_i Y_i = \hat{b}_0 \sum X_i + \hat{b}_1 \sum X_i^2$
(*d*) $\hat{b}_1 = (n \sum X_i Y_i - \sum X_i \sum Y_i)/[n \sum X_i^2 - (\sum X_i)^2] = \sum x_i y_i / \sum x_i^2$ and $\hat{b}_0 = \overline{Y} - \hat{b}_1 \overline{X}$

6.34 For the data in Table 6.12, find the value of (*a*) $\hat{b}_1$ and (*b*) $\hat{b}_0$. (*c*) Write the equation for the estimated OLS regression line.
Ans. (*a*) $\hat{b}_1 \cong 5.94$ (*b*) $\hat{b}_0 \cong 14.28$ (*c*) $\hat{Y}_i = 14.28 + 5.94X_i$

6.35 (*a*) On a set of axes, plot the data in Table 6.12, plot the estimated OLS regression line in Prob. 6.34, and show the residuals. (*b*) Show algebraically that the regression line goes through point $\overline{X}\overline{Y}$.
Ans. (*a*) See Fig. 6-11 (*b*) At $X_i = 5 = \overline{X}$, $\hat{Y}_i = 14.28 + 5.94(5) = 43.98 \cong \overline{Y} = 44$ (the slight difference due to rounding)

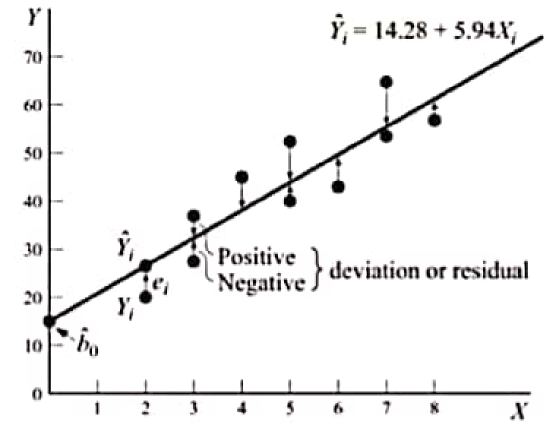

Fig. 6-11

6.36 With reference to the estimated OLS regression line in Prob. 6.34, state (*a*) the meaning of $\hat{b}_0$, (*b*) the meaning of $\hat{b}_1$, and (*c*) the elasticity of Y with respect to X at the means.
Ans. (*a*) $\hat{b}_0$ is the Y intercept (*b*) $\hat{b}_1$ is the slope of the estimated OLS regression line (*c*) $\eta \cong 0.68$

TESTS OF SIGNIFICANCE OF PARAMETER ESTIMATES

6.37 For the data in Table 6.12 in Prob. 6.31, find (*a*) s^2 (*b*) $s_{\hat{b}_0}^2$ and $s_{\hat{b}_0}$, and (*c*) $s_{\hat{b}_1}^2$ and $s_{\hat{b}_1}$.
Ans. (*a*) $s^2 \cong 46.97$ (*b*) $s_{\hat{b}_0}^2 \cong 37.31$ and $s_{\hat{b}_0} \cong 6.11$ (*c*) $s_{\hat{b}_1}^2 \cong 1.31$ and $s_{\hat{b}_1} \cong 1.14$

6.38 Test at the 5% level of significance for (*a*) b_0 and (*b*) b_1 in Prob. 6.34.
Ans. (*a*) b_0 is statistically significant at the 5% level (*b*) b_1 is also statistically significant at the 5% level

6.39 Construct the 95% confidence interval for (*a*) b_0 and (*b*) b_1 in Prob. 6.34.
Ans. (*a*) $0.19 < b_0 < 28.37$ (*b*) $3.31 < b_1 < 8.57$

TEST OF GOODNESS OF FIT AND CORRELATION

6.40 For the estimated OLS regression equation in Prob. 6.34, find (*a*) R^2 and (*b*) r.
Ans. (*a*) $R^2 \cong 0.77$ (*b*) $r \cong 0.88$

6.41 Find the coefficient of rank correlation for the sample of XY observations in Table 6.12.
Ans. $r' \cong 0.90$ ($\cong r \cong 0.88$)

PROPERTIES OF ORDINARY LEAST-SQUARES ESTIMATORS

6.42 With reference to $\hat{b}_0$ and $\hat{b}_1$ in Prob. 6.34, are they (*a*) BLUE? (*b*) Asymptotically unbiased? (*c*) Consistent?
Ans. (*a*) Yes (*b*) Yes (*c*) Yes

6.43 With reference to $\hat{b}_0$ and $\hat{b}_1$ in Prob. 6.34 (*a*) What is the MSE? (*b*) Do $\hat{b}_0$ and $\hat{b}_1$ minimize the MSE?
Ans. (*a*) $\text{MSE}(\hat{b}_0) = \text{var}\,\hat{b}_0$ and $\text{MSE}(\hat{b}_1) = \text{var}\,\hat{b}_1$ (*b*) Yes

SUMMARY PROBLEM

6.44 Table 6.13 gives data for a random sample of 12 couples on the number of children they have Y_i and the number of children they had stated they wanted at the time of their marriage X_i. Regress Y_i on X_i and report your results in summary form.

Ans.
$$\underset{(0.39)}{\hat{Y}_i = 0.22} + \underset{(4.56)}{1.14X_i} \qquad R^2 = 0.68$$

Table 6.13 Number of Children Had and Wanted

Couple	1	2	3	4	5	6	7	8	9	10	11	12
Y_i	4	3	0	4	4	3	0	4	3	1	3	1
X_i	3	3	0	2	2	3	0	3	2	1	3	2

The numbers in parentheses are t values. Thus $\hat{b}_1$ is statistically significant at the 5% (and 1%) level of significance, but $\hat{b}_0$ is not.